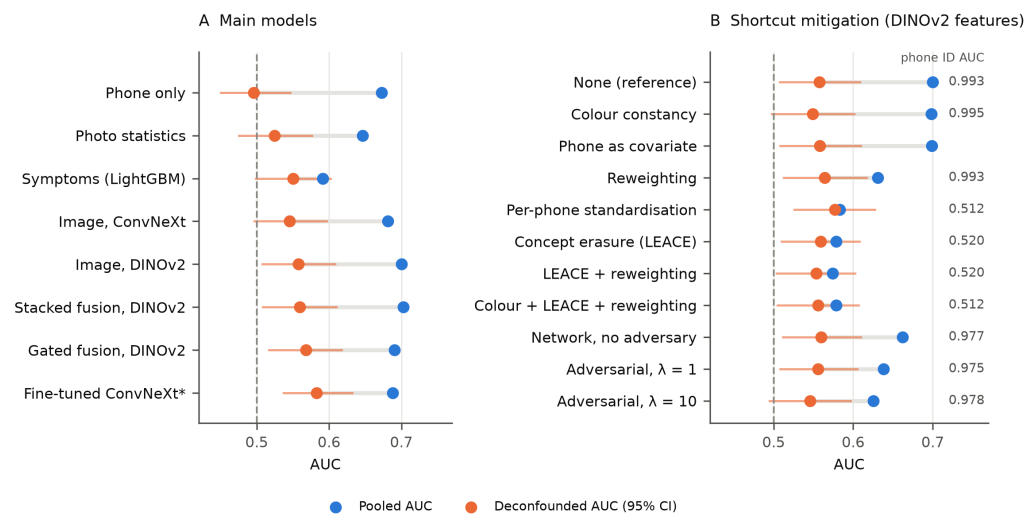


Graphical Abstract

Recognising the phone, not the throat: deconfounded evaluation and shortcut mitigation for smartphone pharyngitis classification

Sandesh Bhatta



Highlights

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- The phone model alone predicts expert pharyngitis labels with AUC 0.67.
- A deconfounded AUC scores phone-only models 0.5 and recovers the shortcut-free AUC.
- Deconfounded, image and fusion models fall from about 0.70 to 0.52–0.58.
- Concept erasure removes the phone signal but recovers no throat signal.
- Adversarial debiasing lowers AUC yet leaves the phone 97–100% identifiable.

Recognising the phone, not the throat: deconfounded evaluation and shortcut mitigation for smartphone pharyngitis classification

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Abstract

Background and Objective: Smartphone throat photographs could help separate bacterial from non-bacterial pharyngitis. On the public PGUPharyngitis dataset (742 patients, two cities, one phone model per city), image models reach AUCs near 0.65–0.70, but bacterial prevalence differs threefold between the phones. We asked how much of this performance comes from recognising the phone, how to measure performance without that shortcut, and whether mitigation methods recover a device-independent signal.

Methods: We analysed 735 patients with physician-consensus labels in repeated, nested 5-fold cross-validation, benchmarking symptom, image and fusion models against phone-only baselines, phone identification and within- and cross-phone evaluation. We introduce a deconfounded AUC that reweights patients so the label is independent of the phone; any phone-only predictor scores 0.5. We validated it by simulation and by resampling the data to set the prevalence gap, and compared it with the covariate-adjusted AUC. We then compared 13 mitigation variants on frozen features (colour constancy, per-phone standardisation, concept erasure (LEACE), reweighting, covariate adjustment, adversarial training) and end-to-end GroupDRO and adversarial fine-tuning, each with a probe of the phone information that remains.

Results: The best pooled AUC was 0.703 (image–symptom fusion); the

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phone alone reached 0.673, and image features identified the phone with AUC 0.99. In simulation the deconfounded AUC was unbiased (bias ≤ 0.001) while pooled AUC was inflated by up to 0.24; on resampled data it stayed within 0.02 of the shortcut-free AUC as pooled AUC rose from 0.54 to 0.78. Deconfounded AUCs of image, symptom and fusion models were 0.52–0.58 (DINOv2 image 0.558, 95% CI 0.51–0.61; phone only 0.496), and the 0.11 advantage of images over symptoms shrank to 0.01 ($p = 0.75$). Standardisation and LEACE removed the phone (probe AUC 0.99 to 0.51–0.52) and lowered pooled AUC to 0.54–0.58, but no method changed deconfounded AUC (all $p > 0.19$). Adversarial training lowered pooled AUC while the phone stayed identifiable (probe AUC 0.97–1.00).

Conclusions: Most apparent image performance on PGUPharyngitis is a device shortcut; the remaining signal (AUC about 0.56) is weak and not improved by current mitigation methods. Deconfounded AUC, site-only baselines and leakage probes, released as an open-source package, should be reported for multi-site imaging models.

Keywords: shortcut learning, confounding, smartphone imaging, pharyngitis, concept erasure, deep learning

1. Introduction

Sore throat is one of the most common reasons for primary-care visits, and most cases are viral. Group A streptococcus and other bacteria cause a minority of cases, yet antibiotics are prescribed far more often than bacterial infection occurs [1, 2]. Clinical scores such as Centor and McIsaac combine fever, tonsillar exudate, tender cervical lymph nodes, absence of cough and age to estimate the probability of streptococcal infection [3, 4]. They are simple and widely taught, but their discrimination is moderate and they depend on a careful physical examination.

Smartphone photographs of the throat are a cheap alternative that also works remotely. Deep-learning models trained on such photographs have detected severe pharyngitis [5] and streptococcal pharyngitis [6] in small single-site studies. In 2025, Shojaei et al. released PGUPharyngitis, the largest public dataset of its kind: 742 throat photographs from patients with cold symptoms in two Iranian cities, taken with two phone models, each labelled bacterial or non-bacterial by four to nine physicians, together with 20 symptoms, age and sex [7]. Their image-only baselines reached an AUC

of 0.645 (DenseNet121), and the dataset was offered to support AI-based diagnosis and remote care. The intended users of such a model are primary-care and telemedicine clinicians deciding whether a sore throat is likely to be bacterial. Streptococcal pharyngitis is most common in school-age children [1], so performance should be checked across age groups.

Two questions remain open. First, the dataset records symptoms, and physicians labelled each case from the photograph and the symptoms together, yet no published model combines the two. Multimodal fusion of images with clinical variables often improves medical prediction [8], so fusion is worth testing here. Second, nobody has checked what image models learn on this dataset. Deep networks readily exploit “shortcuts”: features that predict the label in the training data but are unrelated to the disease [9]. In chest radiography, models identified the hospital and scanner and used them to predict pneumonia or COVID-19 [10, 11], and hip-fracture models leaned on scanner and patient variables [12]. In PGUPharyngitis, each city used a different phone, and bacterial prevalence may differ between cities. A model could therefore score well by recognising the phone.

We set out to build a symptom-aware image model and found that pooled cross-validation on this dataset mostly measures how well a model recognises the phone. Showing that a shortcut exists is only the first step, and it leaves two practical questions for anyone working with multi-site smartphone images. How should a model be scored so that the shortcut earns no credit? And can standard debiasing methods remove the shortcut while keeping a real throat signal? Within-site AUC answers the first question only partly: it discards every comparison between sites and becomes unstable when one site has few positive cases. Mitigation methods from the fairness and domain-adaptation literature (colour constancy, feature harmonisation, concept erasure, reweighting and adversarial training) are rarely compared on the same medical data, and their success is usually judged by pooled accuracy, which a shortcut inflates.

This study makes four contributions:

1. *A shortcut audit of PGUPharyngitis.* The phone alone, or 18 global colour statistics, match the published image baseline; image features identify the phone almost perfectly; image performance largely disappears within a phone and across phones; and the two sites differ in how patients were labelled, not only in prevalence.
2. *A validated deconfounded AUC.* Weighting each patient by $P(y)/P(y |$

site) makes the label independent of the site, so any predictor that uses only the site scores 0.5, while comparisons between patients from different sites are kept. In a simulation with a known answer and on the dataset resampled to different prevalence gaps, it recovered the shortcut-free AUC while pooled AUC rose with the gap; we compare it with the covariate-adjusted AUC of epidemiology. Under this metric the 0.11 AUC advantage of images over symptoms disappears.

3. *A controlled comparison of shortcut mitigation.* Thirteen variants of six method families were tested on two backbones under the same nested protocol, together with end-to-end GroupDRO and adversarial fine-tuning, each with a probe of how much phone information remains. Methods that remove the phone signal lower the pooled AUC to the deconfounded level but do not raise the deconfounded AUC, and adversarial training gives a misleading impression of success.
4. *A multimodal benchmark with open code.* Symptom-only, image-only, clinical-score and five fusion approaches, including a gated cross-modal network, are compared under one repeated, nested cross-validation protocol. Code, folds and out-of-fold predictions are released, and the audit metrics are packaged so that any multi-site study can compute them from its predictions.

2. Related work

Clinical scores. The Centor score assigns one point each for fever, tonsillar exudate, tender anterior cervical lymphadenopathy and absence of cough [3]; McIsaac added an age adjustment [4]. Both are used to decide on testing or empirical antibiotics, and both need findings from a physical examination. PGUPharyngitis records fever and cough but not exudate or lymph nodes, so only a modified score can be computed.

Throat-image AI. Yoo et al. trained convolutional networks on smartphone throat photographs to separate severe pharyngitis from normal throats [5], and Askarian et al. used colour and texture features from smartphone images to detect streptococcal pharyngitis [6]. These studies were small and from single sites. Shojaei et al. released PGUPharyngitis with image-only baselines evaluated by three-fold cross-validation, reporting AUCs from 0.554 (MobileNetV3) to 0.645 (DenseNet121) on the full dataset [7]. To our knowledge, no study has fused these images with symptoms or examined what the models learn.

Multimodal fusion. Medical image–tabular fusion is usually grouped into early fusion (concatenating features), late fusion (combining predictions) and joint or intermediate fusion (learning a shared representation) [8]. Gated units learn how much to trust each modality per sample [13], and feature-wise linear modulation (FiLM) lets one modality scale and shift the features of another [14]. We implement all of these families on equal terms.

Shortcut learning and site confounding. Shortcut learning occurs when a model relies on cues that are predictive in the training distribution but not causally related to the target [9]. In medical imaging, such cues include the hospital, scanner, positioning markers and patient metadata [10, 11, 12]. Hidden stratification, where performance is driven by an easy subgroup, can make overall metrics misleading [15]. Methodological reviews recommend evaluation across sites, explicit confounder baselines and reporting of variance across splits [16]. Brown et al. proposed testing whether a model’s reliance on a sensitive attribute changes its fairness [17], and Ong Ly et al. showed that shortcut learning lowers performance on external data and proposed estimating generalisation without it [18]. We apply these recommendations to throat imaging.

Shortcut mitigation. Methods for removing an unwanted attribute fall into a few families. Colour constancy normalises the illuminant and camera colour cast of each image; Shades of Gray [19] improved dermoscopy classification across acquisition conditions [20]. Feature harmonisation removes site-specific location and scale, as ComBat does for batch effects [21]. Linear concept erasure removes all linearly available information about an attribute; LEACE does so in closed form with the smallest possible change to the features [22]. Reweighting or resampling the training data so that label and nuisance are independent targets the shortcut directly [23]. Adversarial training with a gradient-reversal layer penalises representations from which the domain can be predicted [24], although in text data the removed attribute could still be recovered from the learned representation [25]. We compare these families on the same data and measure how much site information each leaves behind.

Evaluation under confounding. Epidemiology has long adjusted ROC analysis for covariates. The covariate-adjusted ROC curve of Janes and Pepe compares each case only with controls that share its covariate value, and its

area is a weighted average of within-stratum AUCs [26]. Applied to the acquisition site, this removes a site shortcut but also discards every comparison between sites, so a model whose scores are shifted by site is not penalised. Inverse-probability weighting of the evaluation sample keeps those comparisons; we report both.

3. Materials and methods

The study is reported following TRIPOD+AI [27]. All analyses are exploratory: the analysis plan was written before modelling, but hypotheses were not formally frozen before the first results were seen. Figure 1 summarises the study design.

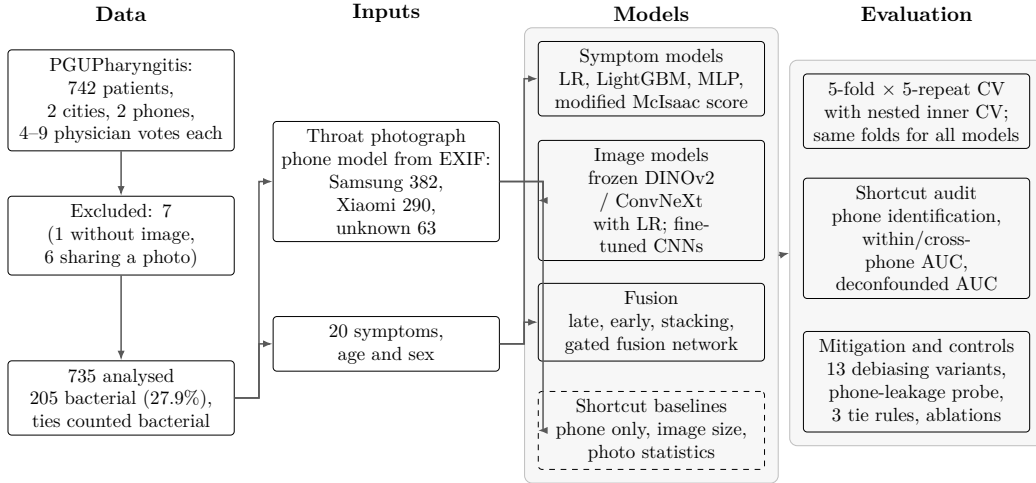


Figure 1: Study design: 742 patients, 735 analysed; inputs (photograph, 20 symptoms, age, sex), model families, 5×5 nested cross-validation, the shortcut audit and the mitigation experiments.

3.1. Dataset

We used PGUPharyngitis (Figshare, <https://doi.org/10.6084/m9.figshare.28163513>, CC BY 4.0) [7]. It contains one throat photograph per patient, taken with a Samsung Galaxy S21 Ultra or a Xiaomi phone in two Iranian cities between October 2023 and May 2024, and a

table with age, sex, 20 binary symptoms and four to nine physician diagnoses (bacterial or non-bacterial). The original study received ethics approval (IR.BPUMS.REC.1403.282) with written informed consent; we used the anonymised public release only.

The table lists 742 patients. Patient 666 had no image. Three pairs of patients (19/23, 218/256, 301/305) shared an identical photograph (perceptual-hash distance 0) but had different ages, sexes or symptoms; because the image could not be assigned to either record, we excluded all six. The analysis set has 735 patients. The table has no city or phone column, so we read the phone model from each photograph’s EXIF metadata: Samsung (model code SM-G998B) for 382 patients, Xiaomi (2201117SG) for 290, and unknown for 63. Because each city used one phone [7], we treat the phone as a proxy for acquisition site and write “phone/site” throughout. Treatments were not recorded. The sample size was fixed by the public dataset and no sample size calculation was done; with 205 bacterial cases, AUC confidence intervals are about ± 0.04 wide, and we report them throughout.

3.2. Labels

The 735 patients received 3,662 diagnoses (median 5 per patient; one patient had 3 recorded). The reference label is the majority vote. One hundred patients (13.6%) had a tied vote; in the primary analysis ties count as bacterial, giving 205 bacterial cases (27.9%). Two sensitivity analyses count ties as non-bacterial (105 bacterial, 14.3%) or exclude them (635 patients, 16.5% bacterial). We also keep the soft label, the share of physicians voting bacterial, and the agreement, $|\text{soft} - 0.5| \times 2$; 230 patients (31%) had unanimous votes. The labels reflect expert consensus from photographs and symptoms, not microbiological testing. The data paper does not report the physicians’ specialties or whether they were blinded to each other’s diagnoses [7].

3.3. Preprocessing

Images were rotated according to EXIF orientation, converted to RGB and resized so the short side was 256 pixels. Training augmentation used a random resized crop to 224×224 (scale 0.8–1.0), horizontal flip, rotation up to 15° and brightness and contrast jitter of 10%. Hue and saturation were not altered, because throat redness is diagnostic. At test time we used a 224-pixel centre crop and averaged predictions over the original and a horizontally flipped image. Images were normalised with ImageNet statistics. Tabular inputs were the 20 symptoms, sex (0/1) and age; age was standardised and

any missing value imputed using statistics from the training data only (no values were missing).

3.4. Evaluation protocol

All models used the same stratified 5-fold cross-validation repeated five times with different seeds (25 test folds of 147 patients), stratified on label $\times$ phone. Fold assignments are released with the code. Within each training part, a 3-fold inner cross-validation selected hyperparameters (by AUC), the number of training epochs for neural fusion models, and the decision threshold, chosen to maximise balanced accuracy on inner out-of-fold predictions. Scalers, imputers, PCA, thresholds and stacking meta-models were fitted on training data only. Because fine-tuning is costly, fine-tuned CNNs used one repeat (5 folds) and a stratified 15% validation split of each training part for early stopping and threshold choice.

3.5. Models

Symptom models. Logistic regression (L2, class-balanced, $C \in \{0.01, 0.1, 1, 10\}$), LightGBM [28] (4 or 8 leaves, 100 or 300 trees, learning rate 0.05) and a two-layer perceptron (64–64 units, L2 penalty $\in \{0.001, 0.1, 1\}$) used the 20 symptoms, sex and age.

Clinical score. A modified McIsaac score summed fever (“Pyrexia”), absence of cough and an age term (+1 for 3–14 years, -1 for ≥ 45 years) [4]; exudate and lymph-node items are not recorded.

Reference baselines. To measure shortcuts we trained three models that do not use throat anatomy: (i) *phone only*, predicting the training-fold bacterial rate of the photograph’s phone; (ii) *image size*, logistic regression on the log width, height, aspect ratio and area of the original photograph; and (iii) *photo statistics*, logistic regression on 18 global statistics of each resized photograph (RGB and HSV means and standard deviations, shares of dark, bright and strongly red pixels, and edge density).

Frozen image models. We extracted embeddings from DINOv2 ViT-S/14 [29] (384 dimensions, 224-pixel input) and ConvNeXt-Tiny pretrained on ImageNet-22k and fine-tuned on ImageNet-1k [30] (768 dimensions), once per image for the test view, the flipped view and ten random training augmentations. Standardised embeddings were classified by class-balanced logistic regression ($C \in \{0.001, 0.01, 0.1, 1\}$).

Fine-tuned image models. To reproduce the published baselines we fine-tuned DenseNet121 [31], MobileNetV3-Large [32] and ConvNeXt-Tiny end to end with AdamW (learning rate 10^{-4} , weight decay 10^{-4} , cosine schedule, batch 16), class-weighted cross-entropy, up to 30 epochs and early stopping on validation AUC (patience 7).

Standard fusion. *Late fusion* averaged the symptom and image logistic-regression probabilities. *Early fusion* fitted one logistic regression on the tabular inputs plus a PCA projection of the image embedding (16 or 64 components). *Stacking* fitted a logistic regression on the logits of inner out-of-fold symptom and image predictions.

Gated cross-modal fusion. Image and symptom embeddings (128 dimensions each) were combined by a learned per-patient gate $g \in (0, 1)$, $z = g h_{\text{img}} + (1 - g) h_{\text{sym}}$, with auxiliary single-modality heads, modality dropout and the physicians’ vote share as a soft target; hyperparameters and the number of epochs were chosen by inner cross-validation. Architecture details and ablations (concatenation, FiLM [14], hard labels, no auxiliary heads, no modality dropout) are in Supplementary Section S4.

3.6. Shortcut audit

1. *Phone identification:* logistic regression predicting Xiaomi versus Samsung (672 patients with known phone) from image embeddings, image size or photo statistics, in 5-fold cross-validation.
2. *Within-phone evaluation:* for each model, AUC computed separately within the Samsung, Xiaomi and unknown-phone strata of the out-of-fold predictions and averaged weighted by stratum size. A phone-only model scores 0.5 by construction.
3. *Within-phone training:* models trained and tested inside one phone (5-fold $\times$ 5 repeats).
4. *Cross-phone transfer:* models trained on all patients of one phone and tested on the other, with patient-level bootstrap confidence intervals (1,000 resamples).
5. *Site differences in patients and labelling:* number of raters, agreement, ties, prevalence under each tie rule and symptom rates, compared between phones (Mann–Whitney and chi-square tests).

Further controls (large photographs only, shuffled images, label dependence) are reported in Supplementary Section S7.

3.7. Deconfounded AUC

Let s_i be the phone group of patient i (Samsung, Xiaomi or unknown) and y_i the label. We give each patient the weight

$$w_i = \frac{\hat{P}(y = y_i)}{\hat{P}(y = y_i \mid s = s_i)}, \quad (1)$$

estimated on the whole cohort. In the weighted cohort every phone group has the same bacterial prevalence, so the label carries no information about the phone. The deconfounded AUC is the weighted AUC,

$$\text{AUC}_w = \frac{\sum_{i:y_i=1} \sum_{j:y_j=0} w_i w_j (\mathbb{I}[p_i > p_j] + \frac{1}{2} \mathbb{I}[p_i = p_j])}{\sum_{i:y_i=1} w_i \sum_{j:y_j=0} w_j}, \quad (2)$$

where p_i is the out-of-fold prediction. A score that depends only on s gets exactly 0.5: pairs within a phone are ties, and for every pair of phones (a, b) the weighted number of bacterial- a /non-bacterial- b pairs equals the number of bacterial- b /non-bacterial- a pairs, so a site offset gains as many pairs as it loses. Unlike within-phone AUC, the metric keeps pairs from different phones and so rewards scores that rank patients consistently across phones. The weights use labels and are an evaluation device only; they are never used to fit the audited models. Because the weights of Eq. (1) are fixed for the cohort, the phone-only model, whose per-phone rates vary slightly across training folds, scores close to but not exactly 0.5. We computed AUC_w per test fold and on the pooled predictions of each repeat, with 1,000 weighted patient-level bootstrap resamples per repeat for confidence intervals.

Estimand and relation to the adjusted AUC. Because the weights change only the joint distribution of label and site and leave the distribution of images within each label–site group unchanged, AUC_w estimates the AUC the same model would reach in a population with the same sites and site sizes in which every site had the overall prevalence. We also report the covariate-adjusted AUC of Janes and Pepe [26], $\text{AUC}_{\text{adj}} = \sum_s \hat{P}(s \mid y = 1) \text{AUC}_s$, the within-site AUCs averaged over the sites of the positive cases. Both give 0.5 to a site-only score. They differ for a model that combines real signal with a site offset: AUC_{adj} ignores the offset, whereas AUC_w counts it against the model, as it would count in a population where the site carries no information.

Validation. We checked both metrics where the answer is known. (i) *Simulation:* two sites (700 patients, 45% at site B, overall prevalence 0.28), a marker with true AUC 0.60, and a site signature that identifies the site (standardised mean difference 3). The prevalence gap between sites was set to 0, 0.1, 0.2, 0.3 or 0.4. In each of 500 replicates per gap, a logistic regression trained on marker and signature in a confounded sample was scored on a confounded test sample with every metric, and on 20,000 patients from the same sites without a prevalence gap; the AUC on this unconfounded sample is the target. The marker alone and the signature alone were scored the same way. (ii) *Resampling PGUPharyngitis:* 200 patients per phone were drawn so that the bacterial rate differed between phones by 0 to 0.4 (overall prevalence 0.25; the full data have a gap of 0.29), 20 times per gap, and each sample was analysed with stratified 5-fold cross-validation of the phone-only, image (DINOv2), image-after-LEACE and symptom models. At a gap of 0 the phone carries no label information, so the image model’s AUC there is a shortcut-free reference.

3.8. Shortcut mitigation

All mitigation methods start from the frozen DINOv2 and ConvNeXt embeddings, use the same outer folds and nested tuning as the reference image model (class-balanced logistic regression with C chosen by inner AUC), and estimate any phone statistics on the training part only:

1. *Colour constancy:* each photograph was corrected with Shades of Gray ($p = 6$) [19, 20] before the embeddings were extracted again.
2. *Phone as covariate:* phone indicators were added to the logistic regression during training and replaced by their training mean at prediction, so the score uses the image only.
3. *Reweighting:* training patients were weighted by Eq. (1) computed on the training part, so the model is fitted to data in which label and phone are independent [23].
4. *Per-phone standardisation:* each embedding dimension was standardised with the mean and SD of the patient’s phone group in the training part, a location-scale harmonisation similar to ComBat without its empirical-Bayes shrinkage [21].
5. *Concept erasure:* LEACE [22] was fitted on the training part to remove all linear information about the phone group (one-hot) from the embeddings, alone or combined with reweighting and colour constancy.

6. *Adversarial training*: a network with one hidden layer (128 units, ReLU, dropout 0.3) was trained with a label head and a phone head behind a gradient-reversal layer [24], with adversarial weight $\lambda \in \{0.3, 1, 3, 10\}$ ramped up over training. The same network with $\lambda = 0$ is the matched control. Training used Adam (learning rate 10^{-3} , weight decay 10^{-3}), 100 epochs, batch 64 and one of the ten cached augmentation embeddings per epoch.
7. *End-to-end fine-tuning*: ConvNeXt-Tiny was fine-tuned with the settings of the fine-tuned baselines (one repeat) using plain class-weighted cross-entropy (ERM), group distributionally robust optimisation over the six phone $\times$ label groups (GroupDRO, step size 0.01) [33], or a gradient-reversal phone head on the pooled features (λ ramped to 1). Early stopping used the deconfounded AUC of the validation split for all three, so that model selection did not reward the shortcut.

To measure how much phone information each method leaves behind, a linear probe (standardised logistic regression, $C = 0.1$) was trained in every outer fold to identify Xiaomi versus Samsung from the transformed training features (for the networks, the hidden layer; for fine-tuned models, the pooled backbone features) and tested on the transformed test features. We also repeated the cross-phone transfer analysis with colour-constancy features and with per-phone standardisation, which needs no labels from the test phone. The adversarial weight was fixed in advance rather than tuned, because tuning on pooled AUC would favour the shortcut.

3.9. Statistics

For each model and repeat, the five test folds were pooled and AUC, PR-AUC, balanced accuracy, macro F1, sensitivity, specificity, Brier score and calibration slope were computed; we report the mean $\pm$ SD across repeats. AUC 95% confidence intervals come from 1,000 patient-level bootstrap resamples within each repeat, averaged across repeats. Model comparisons used the Nadeau–Bengio corrected resampled t -test on the 25 fold-level AUC differences [34], which accounts for overlapping training sets, and the De-Long test [35, 36] within each repeat. Three hypotheses compared gated fusion with the best symptom-only model (H1), the best image-only model (H2) and the best standard fusion model (H3), one-sided, with Holm correction [37]; choosing the best comparator by its own performance makes these tests conservative. Calibration and decision curves [38] are reported in

Supplementary Section S6. Mitigation methods were compared with their reference image model on fold-level pooled, deconfounded and adjusted AUC with the same corrected t -test, two-sided and uncorrected, as exploratory analyses. To check fairness, we report discrimination by sex and age band; no fairness corrections were applied.

3.10. Implementation

Code ran on an Apple M5 MacBook Air (16 GB) with Python 3.11, PyTorch 2.14 (Metal backend), timm 1.0, scikit-learn 1.9 and LightGBM 4.7. SHAP [39], gate and Grad-CAM [40] analyses are described in Supplementary Section S8. The deconfounded and adjusted AUC, the weights of Eq. (1), LEACE and the leakage probe are released as a small Python package (`shortcut_audit`) that takes labels, scores and sites and needs no retraining.

4. Results

4.1. Cohort and differences between phones

The 735 patients had a mean age of 28.8 years (SD 17.1, range 2–83) and 373 (50.7%) were male (Table 1). Bacterial prevalence differed threefold between phones: 14.9% for Samsung and 43.8% for Xiaomi photographs. The phones also differed in how patients were recorded and labelled. Samsung patients were rated by more physicians (median 5 vs 4, $p = 0.006$), physicians agreed more often (mean agreement 0.66 vs 0.47, $p < 10^{-11}$), and tied votes were less common (6.3% vs 23.1%, $p < 10^{-9}$). Because ties were frequent among Xiaomi patients, the tie rule changed the prevalence gap: 29 percentage points with ties counted as bacterial, 12 with ties counted as non-bacterial. Samsung patients reported more symptoms (mean 2.7 vs 1.7, $p < 10^{-19}$; 8 of 20 symptoms differed after Bonferroni correction, Supplementary Table S1) yet had the lower bacterial prevalence. Age and sex did not differ.

4.2. Benchmark

Table 2 summarises the benchmark (ROC curves in Supplementary Fig. S3). The best model was stacked fusion of DINOv2 image and symptom predictions (AUC 0.703, 95% CI 0.66–0.74), followed by the frozen DINOv2 image model alone (0.700). Late, early and gated fusion reached 0.683–0.696. Fine-tuning did not beat frozen features: ConvNeXt-Tiny reached 0.688,

Table 1: Cohort characteristics overall and by phone (63 patients without EXIF phone are included in “All” only). Bacterial counts use the primary rule (ties bacterial) unless stated.

Characteristic	All	Samsung	Xiaomi
Patients	735	382	290
Age, mean (SD), years	28.8 (17.1)	28.8 (17.0)	29.2 (17.6)
Male	373 (50.7%)	204 (53.4%)	142 (49.0%)
Physicians per patient, median (range)	5 (3–9)	5 (4–9)	4 (3–6)
Unanimous votes	230 (31.3%)	154 (40.3%)	61 (21.0%)
Tied votes	100 (13.6%)	24 (6.3%)	67 (23.1%)
Bacterial, ties bacterial (primary)	205 (27.9%)	57 (14.9%)	127 (43.8%)
Bacterial, ties non-bacterial	105 (14.3%)	33 (8.6%)	60 (20.7%)
Symptoms reported, mean	2.27	2.68	1.66

DenseNet121 0.672 and MobileNetV3 0.578, close to the published 0.645 and 0.554 [7]. Symptoms alone were weak (best 0.596), and the modified McIsaac score ranked patients worse than chance (0.437).

Three models that use no throat anatomy did about as well as the published deep networks. The phone-only model reached 0.673, above the published image baseline and above all symptom models. Eighteen global photo statistics reached 0.646, and the original image size 0.562.

4.3. Device and site shortcuts

Every analysis pointed to the phone as the main source of image performance (Table 3, Fig. 2). Image embeddings identified the phone almost perfectly (AUC 0.993 for DINOv2 and 0.997 for ConvNeXt), and so did the 18 photo statistics (0.958). When AUC was computed within each phone, image models fell from about 0.70 to 0.56–0.61, while the phone-only model fell to 0.49, as expected. Models trained and tested inside one phone did worse still: image AUC was 0.50 on Samsung photographs and 0.55–0.58 on Xiaomi photographs. Models did not transfer between phones: trained on one phone and tested on the other, image AUC was 0.50–0.53 and every 95% confidence interval included 0.5. Restricting to large photographs did not change these results (Supplementary Section S7).

The frozen DINOv2 image model beat the phone-only model by 0.025 AUC (corrected t -test, one-sided $p = 0.035$, uncorrected), whereas ConvNeXt did not (+0.006, $p = 0.35$). Within the Xiaomi subgroup the photographs

Table 2: Discrimination under 5×5 cross-validation (fine-tuned CNNs: 1×5). “Within phone” is the size-weighted mean AUC inside each phone stratum. Balanced accuracy and macro F1 use thresholds chosen by inner cross-validation.

Model	AUC [95% CI]	Within phone	Bal. acc.	F1	Brier
Stacked fusion, DINOv2 + sympt.	0.703 [0.66, 0.74]	0.586	0.659	0.637	0.218
Image, frozen DINOv2	0.700 [0.66, 0.74]	0.586	0.656	0.630	0.215
Gated fusion, DINOv2	0.691 [0.65, 0.73]	0.584	0.645	0.613	0.188
Image, fine-tuned ConvNeXt-Tiny	0.688 [0.64, 0.73]	0.605	0.652	0.615	0.211
Gated fusion, ConvNeXt	0.684 [0.64, 0.73]	0.582	0.624	0.606	0.190
Image, frozen ConvNeXt-Tiny	0.681 [0.64, 0.72]	0.561	0.641	0.605	0.218
Phone only	0.673 [0.63, 0.71]	0.488	0.668	0.622	0.182
Image, fine-tuned DenseNet121	0.672 [0.63, 0.71]	0.556	0.622	0.562	0.207
Photo statistics (18 features)	0.646 [0.60, 0.69]	0.536	0.598	0.559	0.233
Symptoms, perceptron	0.596 [0.55, 0.64]	–	0.578	0.563	0.236
Symptoms, LightGBM	0.591 [0.54, 0.64]	0.550	0.561	0.537	0.240
Image, fine-tuned MobileNetV3	0.578 [0.53, 0.62]	0.519	0.550	0.542	0.324
Image size only	0.562 [0.51, 0.61]	0.493	0.536	0.504	0.247
Gated fusion, shuffled images	0.529 [0.48, 0.57]	0.507	0.524	0.500	0.211
Modified McIsaac score	0.437 [0.39, 0.48]	–	0.493	0.245	0.241

Table 3: Shortcut analyses (AUC). In the cross-phone rows, the column names the phone used for testing; the model was trained on the other phone.

Analysis	Samsung	Xiaomi	Overall
Identify phone: DINOv2 / ConvNeXt embeddings	–	–	0.993 / 0.997
Identify phone: photo statistics / image size	–	–	0.958 / 0.754
Within-phone training: DINOv2 image	0.502	0.575	–
Within-phone training: ConvNeXt image	0.499	0.548	–
Within-phone training: DINOv2 + symptoms	0.546	0.561	–
Within-phone training: symptoms	0.627	0.517	–
Within-phone training: photo statistics	0.507	0.538	–
Cross-phone: DINOv2 image [95% CI]	0.502 [0.42, 0.59]	0.519 [0.45, 0.59]	–
Cross-phone: ConvNeXt image [95% CI]	0.507 [0.42, 0.59]	0.530 [0.47, 0.60]	–
Cross-phone: symptoms [95% CI]	0.424 [0.34, 0.52]	0.482 [0.41, 0.55]	–

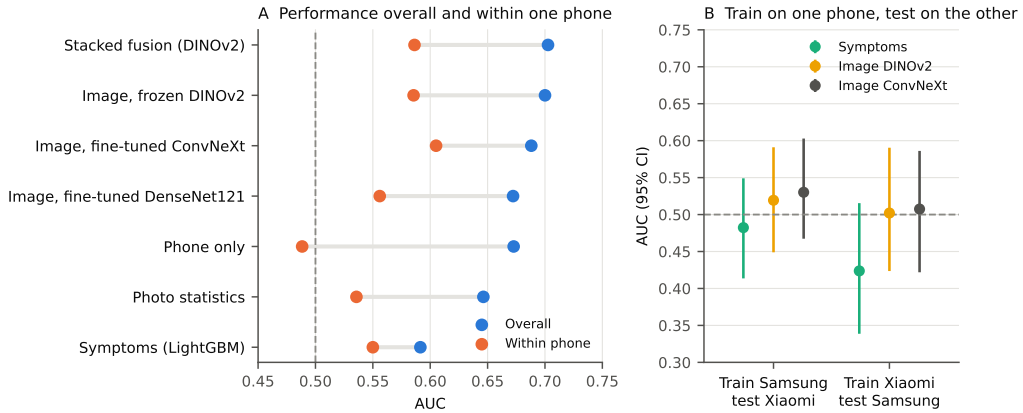


Figure 2: Device and site shortcut. (A) AUC over all patients and the size-weighted AUC within each phone; the phone-only model drops to chance within a phone. (B) Models trained on all photographs from one phone and tested on the other, with 95% bootstrap confidence intervals; all intervals include 0.5.

therefore carry a modest signal (AUC about 0.58–0.61); within the Samsung subgroup they carry essentially none.

4.4. Validation of the deconfounded AUC

In the simulation (Fig. 3A, Supplementary Table S2), the deconfounded AUC matched the target in every setting (largest mean bias 0.001), for the marker alone, the site signature alone and the learned model. The pooled AUC of the learned model rose from 0.60 to 0.75 as the prevalence gap grew, and the site signature alone reached 0.74, although the model’s AUC in an unconfounded population fell from 0.60 to 0.54 because it relied more on the site. The adjusted AUC was unbiased for the marker and the site signature but overestimated the learned model by up to 0.024, because it ignores the site offset in its scores.

On resampled PGUPharyngitis (Fig. 3B, Supplementary Table S3), the image model reached a pooled AUC of 0.535 when the phone carried no label information (gap 0). As the gap grew to 0.4, its pooled AUC rose to 0.775, close to that of the phone alone (0.767), while its deconfounded AUC stayed between 0.533 and 0.552, within 0.02 of the shortcut-free reference. The adjusted AUC stayed between 0.554 and 0.578. After LEACE, pooled AUC stayed between 0.537 and 0.567 at every gap.

4.5. Deconfounded evaluation

Reweightings patients so that the label is independent of the phone removed most of the differences between models (Fig. 4A, Table 4). The phone-only model fell from 0.673 to 0.496 and image size to 0.498. Image, symptom and fusion models all fell to 0.52–0.58, with overlapping confidence intervals whose lower limits ranged from 0.47 to 0.54. The frozen DINOv2 image model kept a deconfounded AUC of 0.558 (95% CI 0.51–0.61), gated fusion 0.568 (0.52–0.62) and the fine-tuned ConvNeXt 0.583 (0.54–0.63, one repeat).

The metric changed several conclusions. On pooled AUC, the DINOv2 image model beat the LightGBM symptom model by 0.111 ($p < 0.001$); on deconfounded AUC the difference was 0.010 ($p = 0.75$). The image model’s lead over the 18 photo statistics shrank from 0.054 ($p = 0.004$) to 0.033 ($p = 0.09$). Conversely, the image model beat the phone-only model by only 0.025 on pooled AUC ($p = 0.07$, two-sided) but by 0.059 on deconfounded AUC ($p = 0.001$), so the photographs do carry some information that the

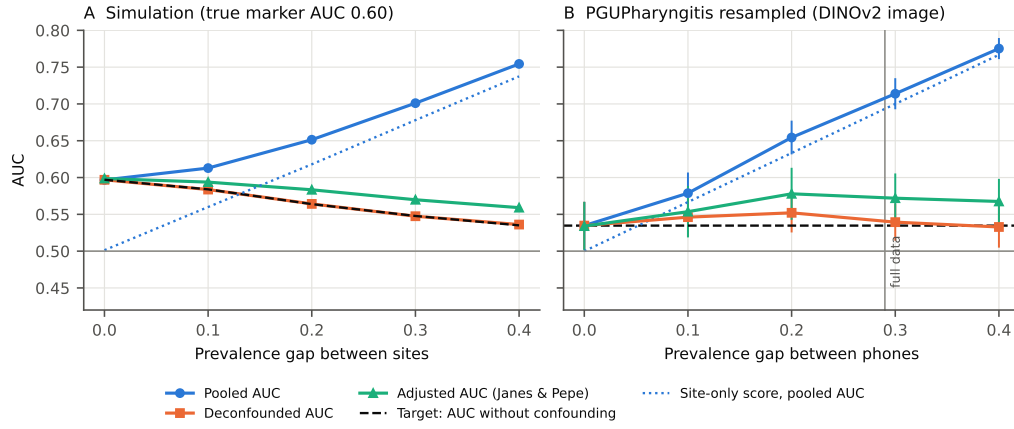


Figure 3: Validation of the deconfounded AUC against known answers. (A) Simulation, 500 replicates per point: logistic regression trained on a marker with true AUC 0.60 and a site signature. The target (dashed) is the same model’s AUC in an unconfounded population from the same sites; the deconfounded AUC lies on it. Dotted: pooled AUC of the site signature alone. (B) PGUPharyngitis resampled to a given prevalence gap between phones (mean $\pm$ SD over 20 resamples, DINOv2 image model). Dashed: the image model’s AUC with no gap; dotted: phone-only pooled AUC; vertical line: the gap in the full data.

phone does not. Fusion did not beat the image alone on either metric (gated DINOv2 +0.013, $p = 0.43$; stacking +0.003, $p = 0.70$).

The adjusted AUC led to the same conclusions: images beat the phone (+0.099, $p < 0.001$) but not the symptoms (+0.043, $p = 0.18$). It was higher than the deconfounded AUC for the image models (DINOv2 0.599 vs 0.558), but not for the symptom models (0.556 vs 0.550) or gated fusion (0.583 vs 0.568). The difference reflects how far a model’s scores are shifted by phone: the image model’s predictions carry a phone offset that the adjusted AUC does not see.

4.6. Shortcut mitigation

Table 5 and Fig. 4B show the mitigation results. Two groups of methods emerged. Per-phone standardisation and LEACE, alone or combined with reweighting and colour constancy, removed the phone information: the probe that identified the phone with AUC 0.99 from the original embeddings fell to 0.51–0.52. Their pooled AUC dropped from 0.70 to 0.57–0.58 with DINOv2 and from 0.68 to 0.54–0.55 with ConvNeXt (all $p < 0.001$), close to the reference model’s deconfounded AUC. Their deconfounded AUC, however, did not change: the largest gain was +0.019 for per-phone standardisation with DINOv2 ($p = 0.22$), and every comparison with the reference had $p > 0.19$. Removing the phone thus removed the inflated part of the score but did not uncover hidden throat signal.

The other methods left the phone identifiable. Colour constancy did not change the probe (0.995 and 0.993) or the pooled AUC, so the phone’s signature in the embeddings is not mainly a colour cast. Phone as covariate left pooled AUC unchanged (0.699 and 0.679), because the image features still encode the phone. Reweighting lowered pooled AUC (0.631 and 0.565) without a significant change in deconfounded AUC. Adversarial training lowered pooled AUC as λ increased (DINOv2: 0.652 at $\lambda = 0.3$ to 0.626 at $\lambda = 10$), which could be read as successful debiasing, yet the hidden layer still identified the phone with AUC 0.97–0.98 at every λ , no better than the matched network without an adversary (0.98). The drop in pooled AUC came from a weaker model, not from removing the phone.

End-to-end fine-tuning gave the same picture (Table 5, bottom). With early stopping on the validation deconfounded AUC, plain ERM reached a pooled AUC of 0.642 and a deconfounded AUC of 0.548. GroupDRO and the adversarial head lowered pooled AUC to 0.609 and reached deconfounded AUCs of 0.550 and 0.567, neither different from ERM (fold-level differences

Table 4: Pooled and deconfounded AUC (Eq. 2) of the main models, mean over five repeats (fine-tuned CNNs: one repeat). The deconfounded 95% CI is from a weighted patient-level bootstrap; the adjusted AUC is the Janes–Pepe covariate-adjusted AUC over the three phone groups. p values compare fold-level AUC with the corrected t -test (two-sided).

Model	Pooled AUC	Deconfounded AUC [95% CI]	Adjusted AUC
Phone only	0.673	0.496 [0.45, 0.55]	0.488
Image size only	0.562	0.498 [0.45, 0.55]	0.482
Photo statistics (18 features)	0.646	0.525 [0.47, 0.58]	0.546
Symptoms, logistic regression	0.574	0.536 [0.48, 0.59]	0.534
Symptoms, LightGBM	0.591	0.550 [0.50, 0.60]	0.556
Image, frozen	0.681	0.545 [0.50, 0.60]	0.563
ConvNeXt-Tiny			
Image, frozen DINOv2	0.700	0.558 [0.51, 0.61]	0.599
Late fusion, DINOv2	0.696	0.559 [0.51, 0.61]	0.589
Early fusion, DINOv2	0.686	0.553 [0.50, 0.61]	0.588
Stacked fusion, DINOv2	0.703	0.559 [0.51, 0.61]	0.599
Gated fusion, DINOv2	0.691	0.568 [0.52, 0.62]	0.583
Gated fusion, ConvNeXt	0.684	0.568 [0.52, 0.62]	0.572
Image, fine-tuned	0.672	0.552 [0.50, 0.61]	0.571
DenseNet121			
Image, fine-tuned	0.578	0.518 [0.47, 0.57]	0.497
MobileNetV3			
Image, fine-tuned	0.688	0.583 [0.54, 0.63]	0.598
ConvNeXt-Tiny			
<i>Paired comparisons: Δ pooled AUC (p) / Δ deconfounded AUC (p)</i>			
Image DINOv2 vs symptoms	+0.111 ($p < 0.001$) / +0.010 ($p = 0.75$)		
LightGBM			
Image DINOv2 vs photo statistics	+0.054 ($p = 0.004$) / +0.033 ($p = 0.09$)		
Image DINOv2 vs phone only	+0.025 ($p = 0.07$) / +0.059 ($p = 0.001$)		
Gated fusion vs image (DINOv2)	−0.005 ($p = 0.72$) / +0.013 ($p = 0.43$)		

+0.015, $p = 0.65$, and +0.032, $p = 0.30$). The phone remained identifiable from the pooled features of every fine-tuned model (probe AUC 0.99–1.00), including the one trained against a phone adversary.

Colour constancy improved cross-phone transfer slightly in all four train–test directions (ConvNeXt: 0.530 to 0.596 and 0.507 to 0.569; DINOv2: 0.519 to 0.537 and 0.502 to 0.535), but the confidence intervals overlapped those of the uncorrected features and only one excluded 0.5 (ConvNeXt, Samsung to Xiaomi, 0.53–0.66). Per-phone standardisation changed cross-phone AUC by less than 0.01.

Table 5: Shortcut mitigation on frozen embeddings (5×5 cross-validation) and with end-to-end fine-tuning (bottom). Pooled and deconfounded AUC are means over repeats; Phone ID is the AUC of a linear probe identifying Xiaomi vs Samsung from the transformed test features (lower means less phone information; 0.5 = none). No method’s deconfounded AUC differed from its reference (corrected t -test, all $p > 0.19$; end-to-end methods were compared with ERM).

Method	DINOv2			ConvNeXt-Tiny		
	Pooled	Deconf.	Phone ID	Pooled	Deconf.	Phone ID
None (reference)	0.700	0.558	0.993	0.681	0.545	0.996
Colour constancy	0.698	0.549	0.995	0.672	0.542	0.993
Phone as covariate	0.699	0.558	–	0.679	0.546	–
Reweighting	0.631	0.564	0.993	0.565	0.518	0.996
Per-phone	0.583	0.577	0.512	0.543	0.547	0.505
standardisation						
LEACE	0.579	0.559	0.520	0.546	0.533	0.516
LEACE + reweighting	0.574	0.553	0.520	0.545	0.532	0.516
Colour + LEACE +	0.579	0.556	0.512	0.535	0.524	0.517
reweighting						
Network, no adversary	0.662	0.560	0.977	0.653	0.566	0.982
Adversarial, $\lambda = 0.3$	0.652	0.560	0.970	0.630	0.559	0.976
Adversarial, $\lambda = 1$	0.638	0.556	0.975	0.612	0.555	0.982
Adversarial, $\lambda = 3$	0.625	0.549	0.975	0.607	0.555	0.981
Adversarial, $\lambda = 10$	0.626	0.546	0.978	0.588	0.557	0.979
<i>End-to-end fine-tuning, ConvNeXt-Tiny (one repeat; early stopping on validation deconfounded AUC)</i>						
ERM (reference)				0.642	0.548	0.994
GroupDRO				0.609	0.550	0.993
Adversarial (gradient reversal)				0.609	0.567	0.998

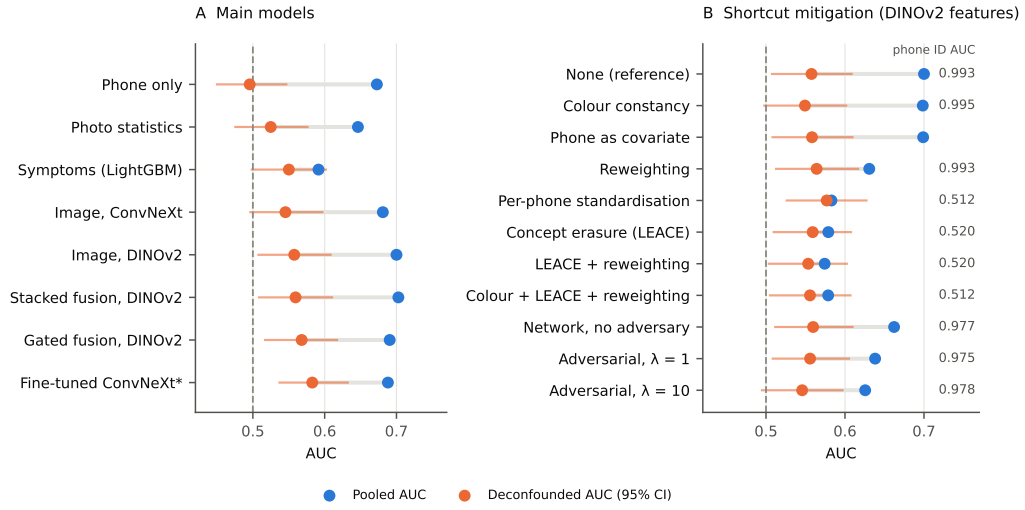


Figure 4: Pooled and deconfounded AUC. (A) Main models: every model that does well on pooled AUC falls to 0.52–0.58 once the label is made independent of the phone; the phone-only model falls to 0.5. *One repeat. (B) Mitigation methods with DINOv2 features; numbers on the right are the phone-identification AUC of a linear probe on the transformed features. Methods that remove the phone lower pooled AUC to the deconfounded level, but none raises the deconfounded AUC.

4.7. Fusion, sensitivity and other analyses

Gated fusion (DINOv2) beat the best symptom model by 0.095 AUC (Holm-adjusted $p = 0.012$) but did not beat the best image-only model (-0.020) or the best standard fusion (-0.008 ; both $p = 1.0$), and its advantage over symptoms was not significant under either alternative tie rule (Supplementary Table S6). No component of the fusion network (gate, auxiliary heads, modality dropout, FiLM) changed AUC by more than 0.012, and a shuffled-image control fell to 0.529, ruling out label leakage (Supplementary Section S4). Under both alternative tie rules, image models stayed at 0.67–0.70 pooled and about 0.61 within phone. Calibration, decision curves (no net benefit over treat-all at thresholds up to 0.20, and less than the phone-only model above that), label dependence, Grad-CAM and subgroup analyses are in Supplementary Sections S6–S8; the image model discriminated better in children (≤ 14 years, AUC 0.77) than in adults ≥ 45 years (0.64).

5. Discussion

We set out to improve smartphone pharyngitis classification by adding symptoms to throat photographs. Instead, the benchmark showed that most of the apparent image performance on PGUPharyngitis comes from recognising the phone, and with it the city, that produced the photograph. The phone alone predicted the expert label as well as the published deep-learning baseline, image embeddings identified the phone almost perfectly, and no model transferred between phones. Two further results go beyond this dataset. A simple reweighting of the evaluation removes the credit a model earns from the site, and under it the differences between image, symptom and fusion models largely vanish. Removing the phone from the features, by several methods that do remove it, does not reveal a stronger throat signal.

Why the shortcut exists. Bacterial prevalence was 44% among Xiaomi photographs and 15% among Samsung photographs. The two phones also differed in who was photographed and how they were labelled: Samsung patients reported more, mostly systemic, symptoms, were rated by more physicians and had far fewer tied votes. Part of the prevalence gap is therefore a property of the labelling rather than of the throats: with ties counted as non-bacterial the gap shrinks from 29 to 12 percentage points. Whether the remainder reflects true epidemiological differences between the two cities,

different referral patterns or different labelling panels cannot be resolved from the released data. Either way, any feature that identifies the phone (colour rendering, sharpness, lighting, framing) becomes predictive of the label. Colour constancy did not reduce phone identification, so the signature is not only a colour cast; our Grad-CAM maps suggest that framing differs too, with confident bacterial predictions on close-ups and confident non-bacterial predictions on wide views. This is the same failure mode reported for chest radiographs, where models recognised hospitals and scanners [10, 11, 12], and it is invisible in pooled cross-validation.

Deconfounded AUC as a reporting standard. Within-site AUC is the usual check for site confounding, but it throws away every comparison between sites and becomes noisy when a site has few positives (57 in our Samsung group). The deconfounded AUC keeps all patients, needs only the site of each patient, and is calibrated against the shortcut: any site-only score gets 0.5. It estimates a well-defined quantity, the model’s AUC in the same sites without a prevalence gap, and both the simulation and the resampling experiment showed that it recovers this quantity while pooled AUC rises with the strength of confounding. The covariate-adjusted AUC of Janes and Pepe [26] is the natural alternative and reached the same qualitative conclusions here. The two answer different questions: the adjusted AUC asks how well a model ranks patients from the same site, the deconfounded AUC how well it ranks patients when the site is uninformative. Only the second penalises a score that is shifted by site, and that shift matters when one threshold is applied across sites or when a model meets a new device. We therefore suggest reporting both, and reading a large gap between them as a sign of a site offset. On PGUPharyngitis the metric changed the ranking that matters for future work: the large pooled advantage of images over symptoms disappeared, while the small advantage of images over the phone became clear. Because it uses standard inverse-probability weights, it can be computed from released out-of-fold predictions without retraining. It answers a narrower question than external validation, since the sites and devices stay the same; it does not remove a shortcut unrelated to the recorded site, and it needs the site to be recorded, which argues for recording it in every dataset.

What mitigation can and cannot do. Methods that removed the phone (per-phone standardisation and LEACE) brought the pooled AUC down to about the deconfounded AUC of the unmitigated model and left the deconfounded

AUC where it was. In this dataset, the device-independent signal that an image model can reach is therefore about 0.55–0.58, whichever way the shortcut is handled, and end-to-end fine-tuning with GroupDRO or an adversarial head did not change this (Table 5). The mitigated models are not better at recognising bacterial pharyngitis, only more honest about how well they do it. This matters in practice, because a debiased model with a pooled AUC of 0.58 is the more trustworthy of the two, even though it looks worse in a table. Adversarial training was misleading: pooled AUC fell steadily with the adversarial weight, which could be reported as successful debiasing, yet a probe recovered the phone as well as from an unadjusted network, both on frozen features and after end-to-end fine-tuning. This agrees with the finding that adversarially removed attributes can be recovered from text representations [25], and it argues for reporting a leakage probe with any debiasing claim. Linear concept erasure is guaranteed to remove linear information [22] and was the simplest effective method here; a nonlinear classifier could still recover some phone information from the erased features, which we did not test.

What remains when the shortcut is removed. Every image, symptom and fusion model kept a deconfounded AUC between 0.52 and 0.58, and the images beat the phone-only model on this metric. The photographs therefore carry some information about the physicians’ judgement beyond the phone, consistent with the residual test, but an AUC below 0.6 is far from clinically useful. The image model also worked best in children, in whom streptococcal pharyngitis is most common, which may deserve a dedicated study.

Fusion and symptoms. Fusion beat symptoms alone only under one of three equally defensible tie rules and never beat the image alone, on any metric, and no component of the fusion network mattered. With symptoms this weak (AUC about 0.6), recorded differently at the two sites, and an image signal dominated by site, there is little for a fusion architecture to exploit: a better architecture will not fix confounded data. The modified McIsaac score ranked patients worse than chance, probably because the dataset lacks exudate and lymph-node items and the physicians were not using a clinical score.

Implications. Smartphone throat-imaging studies, and multi-site smartphone imaging more generally, should (i) report a site- or device-only baseline, (ii) report deconfounded and adjusted AUC alongside pooled AUC,

(iii) evaluate within and across acquisition sites, (iv) report a leakage probe for any debiasing method, (v) balance or randomise devices across sites at data collection and (vi) standardise framing. Datasets should record site and device per image. A deployable model would also need on-device checks of photograph quality and framing, since the models here receive a single photograph and 22 tabular inputs and give no warning when a photograph is unusable; users would need only to take the photograph and record the symptoms. Items (i), (ii) and (iv) need only predictions and site labels; our `shortcut_audit` package computes them in one call, and for PGUPharyngitis our folds and out-of-fold predictions make direct comparison straightforward.

5.1. Limitations

1. Labels are physician consensus from photographs and symptoms, not throat culture or rapid antigen tests, so all results concern agreement with experts rather than infection.
2. City is not recorded; we used the EXIF phone model as a proxy for site and cannot separate phone from city effects. Sixty-three photographs had no EXIF phone.
3. One hundred patients had tied votes; conclusions about fusion versus symptoms depend on the tie rule.
4. The dataset is small (205 bacterial cases) and from one country, and there is no external test set.
5. Fine-tuned CNNs were evaluated with one repeat instead of five, so their estimates are less precise; Grad-CAM maps were not rated by clinicians.
6. The analyses were planned but not formally pre-registered and should be regarded as exploratory.
7. The deconfounded AUC uses weights estimated on the same cohort and removes only confounding by the recorded phone group; it does not replace external validation. Its variance grows when a site has very few cases of one class.
8. Most mitigation methods were applied to frozen embeddings; end-to-end mitigation was tested for one backbone and one repeat. LEACE removes only linear phone information.
9. Labelling differed between sites, so part of the prevalence gap may reflect the labelling panels rather than disease; the data do not allow these to be separated.

6. Conclusion

On the largest public smartphone pharyngitis dataset, image models appear to reach an AUC near 0.70, but most of this performance comes from recognising the phone and site that took the photograph. A deconfounded AUC, under which a site-only model scores 0.5, puts every image, symptom and fusion model between 0.52 and 0.58 and removes the apparent advantage of images over symptoms. In simulation and on resampled data, the deconfounded AUC recovered the shortcut-free AUC while pooled AUC grew with the strength of confounding. Colour constancy, harmonisation, concept erasure, reweighting, GroupDRO and adversarial training did not raise this figure; methods that removed the phone made the pooled AUC honest rather than the model better, and adversarial training only appeared to work. Smartphone imaging models should be judged by deconfounded, within-site and cross-site performance, against a device-only baseline and with a leakage probe for any debiasing claim, before any clinical claim is made.

CRedit authorship contribution statement

Sandesh Bhatta: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics statement

This study used only the anonymised, publicly released PGUPharyngitis dataset. The original data collection was approved by a research ethics committee (IR.BPUMS.REC.1403.282) with written informed consent from

all participants [7]. No additional ethics approval was required for this secondary analysis of public data.

Protocol, registration and patient involvement

No formal study protocol was published and the study was not registered; the analysis decisions are recorded in the code repository (`paper/analysis_plan.md`). Patients and the public were not involved in the design, conduct or reporting of this study.

Data availability

PGUPharyngitis is publicly available under CC BY 4.0 at <https://doi.org/10.6084/m9.figshare.28163513>. Fold assignments and out-of-fold predictions are released with the code.

Code availability

All code, configuration files, fold assignments and out-of-fold predictions are available at <https://github.com/sandesh-bhatta-9/pharyngitis-fusion> under the MIT licence. Five scripts in `scripts/` reproduce every table and figure, and the `shortcut_audit` package computes the deconfounded and adjusted AUC, LEACE and the leakage probe for any study.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to assist with drafting and editing the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Appendix A. Supplementary material

Supplementary material (site differences in symptoms, full validation and mitigation tables, fusion-network details and ablations, planned fusion hypotheses, ROC, calibration and decision curves, additional controls, explainability and subgroup analyses) is provided as a separate file.

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